KAIZEN MATHS

Lesson starter

Quadratic Equations

Non-Standard and Disguised Quadratics

Name _____

S1

$3x - 4 = 10 - 2x^2$

S2

$x^4 - 13x^2 + 36 = 0$

S3

$5x + 4 = 9 - 3x^2$

S4

$(x - 4)^2 - 3(x - 4) - 10 = 0$

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Exit ticket

Quadratic Equations

Non-Standard and Disguised Quadratics

Name _____

E1

$2x - 2 = 8 - 2x^2$

E2

$(x - 4)^2 - 9(x - 4) + 20 = 0$

E3

$4x + 6 = 10 + 2x^2$

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Teacher answers & working

Quadratic Equations · Non-Standard and Disguised Quadratics

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S1

$3x - 4 = 10 - 2x^2$

Answer:

$x_1 \approx 2.00, x_2 \approx -3.50$

Working:

Step 1: Move all terms to one side before applying the quadratic formula.

Step 2: Starting equation: $3x - 4 = 10 - 2x^2$

Move all terms to one side to get standard form $ax^2 + bx + c = 0$.

Step 3: Standard form: $2x^2 + 3x - 14 = 0$

Here a = 2, b = 3, c = -14.

Step 4: Use the quadratic formula: $x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$.

Discriminant $D = 3^2 - 4 \times 2 \times -14 = 121$.

Step 5: Substitute into the quadratic formula:

$x_1 = \frac{-3 + \sqrt{(121)}}{2 \times 2} \approx 2.00$

$x_2 = \frac{-3 - \sqrt{(121)}}{2 \times 2} \approx -3.50$

Step 6: Final answer: $x_1 \approx 2.00, x_2 \approx -3.50$

S2

$x^4 - 13x^2 + 36 = 0$

Answer:

$x = +/-2, x = +/-3$

Working:

Step 1: Use substitution to turn the disguised equation into an ordinary quadratic.

Step 2: Let $y = x^2$. Then the equation becomes:

$y^2 - 13y + 36 = 0$

Step 3: Factor in terms of y: (y - 4)(y - 9) = 0

So $y = 4$ or $y = 9$.

Step 4: Substitute back $y = x^2$:

$x^2 = 4$ or $x^2 = 9$.

$x = +/-2$ or $x = +/-3$

Step 5: Final answer: x = +/-2, x = +/-3

Teacher answers & working

Quadratic Equations · Non-Standard and Disguised Quadratics

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S3

$5x + 4 = 9 - 3x^2$

Answer:

$x_1 \approx 0.70, x_2 \approx -2.37$

Working:

Step 1: Move all terms to one side before applying the quadratic formula.

Step 2: Starting equation: $5x + 4 = 9 - 3x^2$

Move all terms to one side to get standard form $ax^2 + bx + c = 0$.

Step 3: Standard form: $3x^2 + 5x - 5 = 0$

Here a = 3, b = 5, c = -5.

Step 4: Use the quadratic formula: $x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$.

Discriminant $D = 5^2 - 4 \times 3 \times -5 = 85$.

Step 5: Substitute into the quadratic formula:

$x_1 = \frac{-5 + \sqrt{(85)}}{2 \times 3} \approx 0.70$

$x_2 = \frac{-5 - \sqrt{(85)}}{2 \times 3} \approx -2.37$

Step 6: Final answer: $x_1 \approx 0.70, x_2 \approx -2.37$

S4

$(x - 4)^2 - 3(x - 4) - 10 = 0$

Answer:

$x = 9, x = 2$

Working:

Step 1: Use substitution to turn the disguised equation into an ordinary quadratic.

Step 2: Let y = x - 4. Then the equation becomes:

$y^2 - 3y - 10 = 0$

Step 3: Factor in terms of y: (y - 5)(y + 2) = 0

So $y = 5$ or $y = -2$.

Step 4: Substitute back y = x - 4:

$x - 4 = 5$ or $x - 4 = -2$.

Step 5: Solve: x = 9 or x = 2.

Step 6: Final answer: x = 9, x = 2

Teacher answers & working

Quadratic Equations · Non-Standard and Disguised Quadratics

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E1

$2x - 2 = 8 - 2x^2$

Answer:

$x_1 \approx 1.79, x_2 \approx -2.79$

Working:

Step 1: Move all terms to one side before applying the quadratic formula.

Step 2: Starting equation: $2x - 2 = 8 - 2x^2$

Move all terms to one side to get standard form $ax^2 + bx + c = 0$.

Step 3: Standard form: $2x^2 + 2x - 10 = 0$

Here a = 2, b = 2, c = -10.

Step 4: Use the quadratic formula: $x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$.

Discriminant $D = 2^2 - 4 \times 2 \times -10 = 84$.

Step 5: Substitute into the quadratic formula:

$x_1 = \frac{-2 + \sqrt{(84)}}{2 \times 2} \approx 1.79$

$x_2 = \frac{-2 - \sqrt{(84)}}{2 \times 2} \approx -2.79$

Step 6: Final answer: $x_1 \approx 1.79, x_2 \approx -2.79$

E2

$(x - 4)^2 - 9(x - 4) + 20 = 0$

Answer:

$x = 9, x = 8$

Working:

Step 1: Use substitution to turn the disguised equation into an ordinary quadratic.

Step 2: Let y = x - 4. Then the equation becomes:

$y^2 - 9y + 20 = 0$

Step 3: Factor in terms of y: (y - 5)(y - 4) = 0

So $y = 5$ or $y = 4$.

Step 4: Substitute back y = x - 4:

$x - 4 = 5$ or $x - 4 = 4$.

Step 5: Solve: x = 9 or x = 8.

Step 6: Final answer: x = 9, x = 8

Teacher answers & working

Quadratic Equations · Non-Standard and Disguised Quadratics

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E3

$4x + 6 = 10 + 2x^2$

Answer:

No real solutions

Working:

Step 1: Move all terms to one side before applying the quadratic formula.

Step 2: Starting equation: $4x + 6 = 10 + 2x^2$

Move all terms to one side to get standard form $ax^2 + bx + c = 0$.

Step 3: Standard form: $-2x^2 + 4x - 4 = 0$

Here a = -2, b = 4, c = -4.

Step 4: Use the quadratic formula: $x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$.

Discriminant $D = 4^2 - 4 \times -2 \times -4 = -16$.

Step 5: The discriminant is less than 0, so there are no real solutions.

Step 6: Final answer: No real solutions